

Algebra 1 - Q2.1_2.3 B

Name: _____

2.1 – Expressions and Their Parts

1. I can identify the parts of an algebraic expression

In the following expression, list any **terms**, **coefficients**, and **constants**.

$$9x^3 - 2 + 5x$$

Terms:

Coefficients:

Constants:

2. I can describe what a variable, constant, coefficient, and term are

Answer each question about the expression $8x - 3y + 2x^2 - 18$ with **true** or **false**.

- a. The expression has 4 terms _____
- b. The coefficients are 8, -3, and -18 _____
- c. The constant is -18 _____
- d. Both x and y are variables in the expression _____

3. I can write a simple expression to model a real-world situation

Your boss says she will pay you \$25 to show up to work plus \$15 for every hour you work. Write an expression that represents your total pay. Let h be the number of hours you work.

2.2 – Evaluating Expressions

4. I can substitute values into expressions and simplify them

Evaluate the following expressions.

a. $2x - 8$ when $x = -10$

b. $-a^3 + 2b - 3ab$ when $a = 4$ and $b = -2$

5. I can use the correct order of operations with variables and parentheses

Evaluate the following expressions.

a. $3(2x - 1)^2 - x$ when $x = 3$

b. $2|3y - x| + 6y$ when $x = 8$ and $y = -10$

c. $\frac{1}{x}(3x - 2)^2 + x$ when $x = 2$

6. I can use substitution to test whether two expressions are equivalent

Test whether $9x - 3$ is equivalent to $3(x - 1) + 6x$. Try **2 different values** for x .

2.3 – Combining Like Terms

7. I can identify like terms in an expression

For each of the following, indicate whether the terms are like terms (**true** or **false**).

a. $-5x$ and $-5y$ _____

b. $-x^3$ and $2x^3$ _____

c. $-5xy$ and $3yx$ _____

d. $5x^2y^2$ and $-9x^2y^2$ _____

8. I can combine like terms to simplify an expression

Simplify the following expressions by combining like terms.

a. $22 - 8x^2 + 3x - 11x + 21$

b. $2a + b - 3a + 5b^2 - 2 + 9b^2 - 1$

9. I can check my work by substitution or by reasoning about equivalence

Simplify the following expression

$$-x - 5y^2 + 3x - 2 - 6y$$

Test whether your answer is equivalent to the original expression.